

ezr, week 1: columns, streaming, forgetting

the story

One weekly-sized chunk of `ezr.lua`, verbatim, with glossary notes folded in and this week's exercises at the bottom.

A column watches values stream past and keeps a tiny summary: a `Num` holds mean and standard deviation, a `Sym` holds counts, mode and entropy. Nothing stores the values themselves. The stats also run backwards: subtract a value and they roll back, so a table can forget rows as cheaply as it learned them.

noir I

Nominal, Ordinal, Interval, Ratio (Stevens 1946): the four scales of measurement. This code collapses them to two: symbols you can only count (nominal) and numbers you can subtract (interval and up). One header letter decides which:

```
function Col(name,at)
  return (name:find"%l" and Sym or Num)(name,at) end
```

Num, Sym I

The summary of a numeric column: count `n`, mean `mu`, and `m2` (the sum of squared deviations from the mean, from which the standard deviation falls out). Nothing else is stored — not the data, just three numbers. A trailing `-` in the name means “goal: minimize”.

That last line, in Python, uses the True/False-is-1/0 trick (bools ARE ints, so arithmetic on a test needs no if):

```
heaven = 1 - (name[-1] == "-") # True==1, False==0
```

The summary of a symbolic column: count `n` and a table of counts `has`. Again, no data kept — just the histogram.

```
function Num(name,at)
  name = name or ""
  return new(NUM, {at=at or 1, name=name, n=0, mu=0, m2=0,
                  heaven = name:find"~$" and 0 or 1}) end

function Sym(name,at)
  return new(SYM, {at=at or 1, name=name or "", n=0, has={}}) end
```

columnProtocol, welford I

Num and Sym answer the same eight questions — one polymorphic protocol, two implementations. Everything downstream (tables, distance, trees, cuts) talks to the protocol, never to the type:

question	Num answers	Sym answers
add	welford update of mu, m2	bump a count in has
sub	welford, run backwards	drop a count
mid	mean	mode
div	standard deviation	entropy
norm	cdf position, 0..1	identity
dist	gap of normed values	0 if same else 1
holds	$x \leq v$	$x == v$
reset	zero mu, m2	empty has

Two samples, both sides of add:

Note the shared conventions: "?" (missing) is ignored on the way in, and `inc=-1` runs the summary backwards (see stream).

Welford's 1962 one-pass update: mean and variance from a stream, no stored data, no catastrophic cancellation. After each value v :

$$n' = n + 1, \quad d = v - \mu, \quad \mu' = \mu + d/n', \quad m_2' = m_2 + d(v - \mu')$$

then $sd = \sqrt{m_2/(n-1)}$. The NUM.add code above is exactly these four lines. Run with `inc=-1` the algebra inverts, which is what makes summaries subtractable.

```
function SYM.add(i,v,inc)
  if v == "?" then return v end
  inc = inc or 1
  i.n = i.n + inc
  i.has[v] = inc + (i.has[v] or 0)
  if i.has[v] <= 0 then i.has[v] = nil end
  return v end

function NUM.add(i,v,inc, d)
  if v == "?" then return v end
  inc = inc or 1
  i.n = i.n + inc
  d = v - i.mu
  i.mu = i.mu + inc * d / i.n
  i.m2 = i.m2 + inc * d * (v - i.mu); return v end
```

mid (mode, mean), diversity (entropy, standard deviation) I

The most frequent symbol: a Sym's answer to mid ("what is typical here?"). The **mean is the same question asked of numbers** — both are one value standing in for the whole column. That is why mid is one protocol slot, not two functions with different names:

Shannon 1948: the spread of a symbol column, in bits — the mean surprise of drawing from counts $p_k = n_k/n$:

$$e = - \sum_k p_k \log_2 p_k$$

All-same symbols: 0 bits. Uniform over k symbols: $\log_2 k$ bits. **Variance (or sd) is the same question asked of numbers** — "how far is this column from settled?" — which is why div ("diversity") is one protocol slot with two spellings:

The two analogies are one design rule: every protocol slot names a question; each type answers in its own dialect. Central tendency: mean, mode. Diversity: sd, entropy. Trees built on div therefore handle numeric and symbolic goals with the same code.

```
function SYM.mid(i, hi,out)
  hi = -1
  for k, n in pairs(i.has) do
    if n > hi then hi, out = n, k end end
  return out end

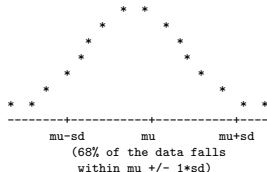
function NUM.mid(i) return i.mu end

function SYM.div(i)
  return sum(i.has, function(n, p)
    p = n / i.n
    return -p * log(p) / log(2) end) end

function NUM.div(i)
  return i.n < 2 and 0 or sqrt(max(i.m2,0) / (i.n-1)) end
```

gaussian (mean, second moment), cdf (cumulative distribution function) |

The bell curve:



Summarized by two moments: the first moment μ (the mean, `mu`) and, from the second moment, the spread (`m2` = sum of squared deviations, so $sd = \sqrt{m2/(n-1)}$). Two tricks make these course-critical: both update *incrementally*, one value at a time (`welford`); and two summaries *subtract* without resampling. From `ezr-eg1`'s --without: pour `{10,20,30}` into a summary of `{1,2,3,4,5}`, subtract a summary of `{10,20,30}`, and `mu` and `sd` of `{1..5}` come back exactly. Learn, unlearn, in $O(1)$ — no stored data (see stream). The fraction of a population at or below a value: the area under the pdf up to x . Monotone, 0..1 — which makes it a natural normalizer: `norm` maps any cell to “what fraction of this column sits below you?”. The normal cdf has no closed form, so `ezr` uses a logistic approximation (good to about $\pm 1\%$), with the z -score clamped to ± 3 :

$$cdf(z) \approx \frac{1}{1 + e^{-1.702 z}}$$

A bonus, for later: discretization rides on this for free. To turn any number into a small bin id, take `floor(the.bins * num:norm(23))` — since `norm` is the cdf, equal-width slices of 0..1 give (roughly) equal-frequency bins of the data.

```
function NUM.norm(i,v, z)
  if v == "?" then return v end
  z = (v - i.mu) / (i:div() + TINY)
  return 1 / (1 + exp(-1.702 * max(-3, min(3, z)))) end
```

stream l

A summary you can update — and un-update — one datum at a time, in constant memory. The `inc` argument (+1 or -1) means adding is $O(1)$ and so is deleting, so any add-and-forget sweep over n items runs in linear time. Remember that: it matters later. Trees will score EVERY possible split of a sorted column in one linear pass — adding each row to the summary on one side of the cut while forgetting it from the other — where naive rebuilding would cost $O(n^2)$. It is also why $(a+b)-b == a$ is a testable law (--without, --sub in `ezr-egl`).

```
function NUM.__sub(i,j,    n,d)
  n = i.n - j.n
  if n < 1 then return Num(i.name, i.at) end
  d = j.mu - i.mu
  return new(NUM, {name=i.name, at=i.at, heaven=i.heaven,
                  n=n, mu=(i.n*i.mu - j.n*j.mu) / n,
                  m2=max(0, i.m2 - j.m2
                        - d*d*i.n*j.n/n)}) end

function NUM.reset(i) i.n, i.mu, i.m2 = 0, 0, 0 end
function SYM.reset(i) i.n, i.has = 0, {} end
```

exercises I

Exercises, week 1

1. Port this page's code to Python, wiring each demo of `ezr-eg1` to a `test_` function.
2. Add $\{10, 20, 30\}$ to a Num one value at a time, printing `mu` and `sd` after each add. Watch Welford converge.
3. What is the entropy of a Sym fed "a", "a", "a"? Fed "a", "b", "c"? Predict first, then run it.
4. In `NUM.__sub`, why does `m2` get clamped with `max(0, ...)`? What does that say about floats?
5. Flip `heaven` to and 1 or 0. Every test still passes. What breaks, and how would you ever notice?